

SUPERIOR SHOALS, NY, USA

WMO: 994580

Lat: 44.4645N

Lon: 75.7936W

Elev: 254

StdP: 14.56

Time Zone: -5.00 (NAE)

Period: 94-09

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.0	-0.1	-17.3	2.2	-4.1	-11.3	3.0	1.6	47.0	38.7	40.2	31.8	8.6	40	0.630

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	81.3	70.4	79.3	69.4	77.4	68.4	73.4	77.9	72.2	76.5	70.9	75.2	12.2	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.1	119.7	75.7	70.7	114.1	75.0	69.2	108.4	74.1	37.2	78.0	36.1	76.7	34.9	75.1	79.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
38.7	32.0	26.2	DB	-9.4	85.1	7.1	2.1	-14.5	86.7	-18.6	87.9	-22.6	89.1	-27.8	90.7	
			WB	-9.7	76.1	7.6	2.1	-15.2	77.6	-19.7	78.8	-24.0	80.0	-29.5	81.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.5	20.8	22.5	31.3	43.3	54.8	64.8	69.6	69.0	62.3	50.2	39.8	28.1	
	DBStd	19.03	12.34	10.15	9.62	8.02	6.06	5.58	4.28	4.65	6.42	7.60	8.46	9.86	
	HDD50	3604	904	771	581	229	23	0	0	0	2	93	320	681	
	HDD65	7156	1369	1191	1044	650	320	71	10	16	124	460	756	1145	
	CDD50	2330	0	0	2	29	171	444	607	590	371	101	14	1	
	CDD65	405	0	0	0	0	3	65	152	141	42	2	0	0	
	CDH74	1383	0	0	0	0	16	210	512	505	139	1	0	0	
	CDH80	113	0	0	0	0	0	20	36	44	13	0	0	0	
Wind	WSAvg	11.6	12.8	13.0	12.9	12.6	11.0	9.4	10.1	9.0	10.7	12.4	13.0	12.9	
Precipitation	PrecAvg	38.7	2.9	2.4	2.6	3.1	3.3	3.2	3.6	3.1	3.6	4.1	3.6	3.3	
	PrecMax	48.3	6.1	4.2	4.8	5.7	5.7	6.9	7.8	4.6	6.8	7.2	5.7	5.6	
	PrecMin	30.8	1.0	1.0	1.1	0.4	1.3	0.7	1.3	0.9	1.9	0.7	1.7	1.5	
	PrecStd	4.2	1.3	0.7	1.0	1.4	1.3	1.5	1.5	1.2	1.2	1.8	1.0	1.1	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	48.3	61.0	69.9	76.5	82.6	83.2	83.8	81.9	73.0	63.1	55.5	
		MCWB	51.5	43.4	50.8	55.6	63.8	69.2	72.4	71.2	68.9	62.9	55.3	51.4	
	2%	DB	45.9	44.0	53.4	64.6	71.5	79.0	80.9	81.1	77.9	68.7	59.7	49.0	
		MCWB	43.9	40.5	45.1	52.7	60.3	68.5	71.2	70.4	67.5	61.6	54.1	46.4	
	5%	DB	41.2	39.9	48.6	59.9	68.1	76.4	78.9	79.0	74.7	65.4	56.3	45.1	
		MCWB	38.7	36.7	42.9	50.6	57.8	67.1	69.7	69.3	66.5	59.2	51.7	42.4	
	10%	DB	37.1	36.6	44.2	55.3	64.9	73.6	77.0	77.0	71.9	61.9	51.9	41.2	
		MCWB	34.9	33.7	39.4	47.5	56.9	66.0	68.7	68.8	64.9	56.3	47.7	38.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	45.2	53.3	59.8	68.3	73.0	75.4	76.1	72.4	66.3	58.7	52.3
			MCDB	54.3	46.6	59.3	64.6	72.5	78.2	79.9	79.9	78.1	70.9	61.3	55.2
2%		WB	44.3	40.9	47.4	55.7	63.9	71.1	73.7	73.3	69.8	62.2	55.4	46.8	
		MCDB	46.0	42.8	50.6	61.4	69.0	76.0	78.2	77.5	74.5	66.8	58.5	48.6	
5%		WB	38.8	37.0	43.7	52.0	61.1	69.2	72.5	72.1	68.4	59.7	52.1	42.8	
		MCDB	40.9	39.8	48.2	57.8	65.4	73.8	76.8	76.1	73.0	64.1	55.9	45.1	
10%		WB	35.2	33.6	40.1	48.3	58.7	67.7	71.0	70.8	66.5	57.1	48.6	39.0	
		MCDB	37.0	35.6	44.0	55.0	63.0	72.4	75.2	74.9	71.2	61.5	51.6	41.2	
Mean Daily Temperature Range	5% DB	MDBR	13.4	14.7	13.5	14.0	14.2	13.0	11.5	12.1	13.1	11.5	10.5	11.3	
		MCDBR	15.7	16.6	19.9	21.2	19.5	15.8	13.5	13.5	15.7	15.7	17.1	14.6	
		MCWBR	14.7	14.6	14.1	12.4	10.2	8.1	7.8	7.4	8.7	10.8	14.1	13.0	
	5% WB	MCDBR	15.9	16.5	19.6	18.8	16.8	13.4	11.5	11.2	13.5	14.4	16.1	14.7	
		MCWBR	15.1	14.7	15.3	13.5	11.2	8.1	8.1	7.2	8.5	10.9	14.1	13.8	
Clear-Sky Solar Irradiance	taub	0.290	0.295	0.324	0.354	0.383	0.417	0.429	0.418	0.368	0.349	0.310	0.285		
	taud	2.373	2.359	2.417	2.395	2.357	2.277	2.249	2.281	2.408	2.456	2.508	2.452		
	Ebn at Noon	264	284	290	288	281	271	266	265	271	261	256	255		
	Edh at Noon	24	29	32	36	38	42	42	40	32	27	22	20		
All-Sky Solar Radiation	RadAvg	427	709	1128	1398	1675	1792	1835	1616	1286	763	479	329		
	RadStd	42	72	92	92	170	173	143	158	75	109	73	52	40	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page